Group I, Row V, part 2b: the UV divergence in dimensional regularization ($\overline{\rm MS}$)

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1 Result

Part 2b, as you wrote it, has one UV divergence. It is at $x \to z$, where the two gluons of the loop (at x with p^+, at z with $k^+ - p^+$) come together. We add and subtract the short-distance limit of the integrand (the counterterm, Sec. 4).

- The counterterm is integrated over x in $d_\perp = 2-\epsilon$ dimensions in $\overline{\rm MS}$, and over p^+ exactly. This gives the **UV term**.
- What remains is the **rest**. It is UV finite and is evaluated in two dimensions.

Nothing is expanded in Λ/k^+ or $k^+/\vee$.

$$\boxed{\frac{d^3N_{2b}}{d^3k} = \frac{d^3N_{2b}}{d^3k}\bigg|_{\rm UV} + \frac{d^3N_{2b}}{d^3k}\bigg|_{\rm rest} + \text{c.c.}} \tag{1.1}$$

The UV term (dimensional regularization, $d_\perp = 2-\epsilon$, $\overline{\rm MS}$ scale μ).

$$\begin{aligned}\frac{d^3N_{2b}}{d^3k}\bigg|_{\rm UV} = \frac{1}{(2\pi)^3}\,\frac{g^4N_c}{16\pi^4}\,\frac{1}{k^+}\left[2\log\left(\frac{\vee}{\Lambda}\right) - 2\log\left(\frac{\vee}{k^+}\right) + 2\log\left(1-\frac{\Lambda}{k^+}\right) - \frac{11}{6} + \frac{4\Lambda}{k^+} - \frac{\Lambda^2}{k^{+2}} + \frac{2\Lambda^3}{3k^{+3}}\right]\\ \times \int_{y,z,x',w'} e^{-ik\cdot(w'-z)}\left[\frac{2}{\epsilon} + 2\gamma_E + \log\frac{(y-z)^2\mu^2_{\overline{MS}}}{4}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\\ \times \Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[U^{af}(y) - U^{af}(z)\Big]\rho^{b'}(x')\,\rho^f(y).\end{aligned} \tag{1.2}$$

The bracket with the pole comes from the x integral of the counterterm (Sec. 5). In transverse coordinate space it is

$$\frac{2}{\epsilon} + 2\gamma_E + \log\frac{(y-z)^2\mu^2_{\overline{MS}}}{4} = \frac{1}{\pi}\int_x\Big[K(z,z,x) - K(y,z,x)\Big] = \frac{1}{\pi}\int_x \frac{(y-x)\cdot(y-z)}{(y-x)^2\,(x-z)^2} \qquad (d_\perp = 2-\epsilon), \tag{1.3}$$

with $K(a,b,c) = \frac{(a-c)\cdot(b-c)}{(a-c)^2(b-c)^2}$. This is the same form as in your G1-II, $\frac{1}{\pi}\int_z[K(x,w,z) - K(w,w,z)] = -\big[\frac{2}{\epsilon} + 2\gamma + \log\frac{(x-w)^2\mu^2_{\overline{MS}}}{4}\big]$.

After the $\overline{\rm MS}$ subtraction (remove $\frac{2}{\epsilon}$; the γ_E and $\log 4\pi$ of $\overline{\rm MS}$ are already in $\mu_{\overline{MS}}$):

$$\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y-z)^2\mu^2_{\overline{MS}}}{4}\;\longrightarrow\;\log\frac{(y-z)^2\mu^2_{\overline{MS}}\,e^{2\gamma_E}}{4}\quad\text{in Eq. (1.2).}\tag{1.4}$$

The rest (UV finite; $d_\perp=2$; no plus prescription).

$$\begin{aligned}
\left.\frac{d^3N_{2b}}{d^3k}\right|_{\rm rest} &= \frac{1}{(2\pi)^3}\frac{g^4}{8\pi^4}\frac{1}{k^+}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\Bigg\{\\
&e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{1}{[p^+(y-x)^2+(k^+-p^+)(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\\
&\times\Bigg[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\left[(y-x)^2-(y-z)^2\right]\frac{p^+(k^+-p^+)}{k^{+2}}-\frac{p^+}{k^+-p^+}\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2}-\frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right]\\
&\quad+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2(y-x)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right]\\
&\quad-\frac{k^+-p^+}{p^+}\left[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right]\\
&\quad+\frac{p^+\,\delta_{im}}{k^+}\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2(y-z)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right]\\
&\quad-\left[\frac{(y-x)^m(x-z)^i}{(x-z)^2}-\frac{(y-x)^m(y-z)^i}{2(y-z)^2}+\frac{(y-z)^m(x-z)^i}{(x-z)^2}+\frac{(y-x)^i(y-z)^m}{2(y-x)^2}\right]\Bigg]\\
&\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{k^+}\left[\frac{p^+(k^+-p^+)}{k^{+2}}+\frac{p^+}{k^+-p^+}+\frac{k^+-p^+}{p^+}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{(y-x)\cdot(y-z)}{(y-x)^2\,(x-z)^2}\\
&\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(z)\Big]\rho^{b'}(x')\,\rho^f(y)\Bigg\}.
\end{aligned}\tag{1.5}$$

Everything before the last two lines is your expression for 2b, unchanged (in the rest $d_\perp=2$). The last two lines subtract the counterterm.

Notes.

1. **Conventions.** $d_\perp=2-\epsilon$, as in your G1-II, so $\frac{2}{\epsilon}=\frac{1}{\epsilon_{\rm std}}$ in the usual $D=4-2\epsilon_{\rm std}$ notation. Every $d_\perp$-dimensional transverse integral carries $\left(\frac{\mu^2e^{\gamma_E}}{4\pi}\right)^{\epsilon/2}$, $\mu\equiv\mu_{\overline{MS}}$. In $\overline{\rm MS}$ only $\frac{2}{\epsilon}$ is then subtracted.

2. **The log rule.** The p^+ integral gives $2\log\frac{k^+-\Lambda}{\Lambda}$. We used $\log\frac{k^+}{\Lambda}=\log\frac{\vee}{\Lambda}-\log\frac{\vee}{k^+}$, i.e. your $\log\frac{\Lambda}{k^+}=-\log\frac{\vee}{\Lambda}+\log\frac{\vee}{k^+}$, so $\log\frac{k^+-\Lambda}{\Lambda}=\log\frac{\vee}{\Lambda}-\log\frac{\vee}{k^+}+\log\left(1-\frac{\Lambda}{k^+}\right)$.

3. **The UV term is LO times a factor.** Its prefactor is $\frac{\alpha_sN_c}{2\pi}$ times that of LO. The integral is the LO cross section with $x\to y$ (the charge) and $w\to z$ (the gluon), with the bracket of Eq. (1.3) inside the integral.

4. **Running coupling.** The $-\frac{11}{6}$ term is $-\frac{\alpha_s\beta_0}{4\pi}\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y-z)^2\mu^2}{4}\right]\times$LO, with $\beta_0=\frac{11}{3}N_c$. Your G1-II (part 1) has the same term with the opposite sign, $+\frac{\alpha_s\beta_0}{4\pi}\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(x-w)^2\mu^2}{4}\right]\times$LO. After renaming $y\to x$, $z\to w$, the two cancel exactly per ordering, pole and logarithm included (Sec. 7.3).

5. **The pole multiplying the rapidity logarithms** is not a renormalization of the coupling. In the form of Eq. (1.3), it is an x integral with a $1/(x-z)^2$ singularity, the same type as your G1-II after F6. These terms are part of the $\log(\vee/\Lambda)$ two-ρ terms. Their sum is JIMWLK$\otimes$LO, where such singularities cancel between rows (Sec. 7.4).

6. **Choice of counterterm.** The counterterm is cut off at the distance $|y-z|$ through $K(z,z,x)-K(y,z,x)$. The cut $\theta(|y-z|-|x-z|)/(x-z)^2$ and the smooth form $\frac{1}{(x-z)^2}-\frac{1}{(x-z)^2+(y-z)^2}$ give the same UV term. They also give the same rest after the x integration (Sec. 4).

2 Notation and scheme

- $\xi \equiv p^+/k^+$, $\mathbf{r} \equiv x - z$, $Y \equiv y - z$; $\mathcal{K}(X) \equiv X/X^2$.
- $K(a,b,c) \equiv \frac{(a-c)\cdot(b-c)}{(a-c)^2(b-c)^2}$, as in your Appendix E: the third argument is the point that is integrated over. So $K(z,z,x) = \frac{1}{(x-z)^2}$, and $K(y,z,x) = \frac{(y-x)\cdot(z-x)}{(y-x)^2(z-x)^2}$.
- The phase $e^{-ik\cdot(z-x)p^+/k^+}$ of your expression depends on x, z and p^+, so it stands inside $\int_{x,z}$ and $\int dp^+$. We write all transverse integrals together, $\int_{x,y,z,x',w'}$.
- **Scheme** (the same as for G1-II):
  - the loop integral, over the size $x - z$ of the gluon pair, is done in $d_\perp = d = 2 - \epsilon$ at fixed y, z, x', w';
  - the gluon polarizations are d-dimensional (your $d_\perp$ in the first term of the bracket);
  - the external factors stay two-dimensional.
- $\alpha_s = \frac{g^2}{4\pi}$; $\beta_0 = \frac{11}{3}N_c$ (gluons only).
- LO (`main.tex`):

$$\frac{d^3N_{\rm LO}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2k^+}\int_{x,w,x',w'} e^{-ik\cdot(w'-w)}\frac{(x-w)\cdot(x'-w')}{(x-w)^2(x'-w')^2}[U(x)-U(w)]^{ab}[U(x')-U(w')]^{ab'}\rho^b(x)\rho^{b'}(x').$$

3 Where part 2b is UV divergent

3.1 The regions

- $x \to z$: the integrand behaves like $1/(x-z)^2$, and the colour factor $N_c[U(y)-U(x)] \to N_c[U(y)-U(z)]$ does not vanish. $\int d^2x/(x-z)^2$ diverges logarithmically. This is the UV divergence.
- $y \to x$ and $y \to z$: at most $1/|y-x|$ or $1/|y-z|$, which is integrable.
- x, y, z all together: the integrand scales as ρ^{-3} against $d^2x\,d^2y \sim \rho^3 d\rho$, and the colour factor vanishes as well. Integrable.
- Large $|x-z|$: the integrand falls like $1/(x-z)^2$. Its angular average is not zero, so the x integral converges because of the phase $e^{-ik\cdot(z-x)p^+/k^+}$ (checked numerically, Table 1). This holds for your expression as written, and the subtraction below does not change it: the counterterm falls like $|x-z|^{-3}$.

3.2 The short-distance limit

Put z at the origin, $x = \mathbf{r}$, $y = Y$, and let $r \to 0$. Then:

- $e^{-ik\cdot(w'-z)}e^{-ik\cdot(z-x)\xi} \to e^{-ik\cdot(w'-z)}$;
- $\frac{1}{p^+(y-x)^2+(k^+-p^+)(y-z)^2} \to \frac{1}{k^+(y-z)^2}$;
- $N_c[U(y)-U(x)] \to N_c[U(y)-U(z)]$.

Call the six terms of your bracket $B_1, \ldots, B_6$ in the order written, without their p^+ factors. Contracted with $\frac{(z-x)^m}{(z-x)^2} = -\frac{r^m}{r^2}$, their leading terms are:

term	p^+ factor	leading term of $-\frac{r^m}{r^2}B^{im}$	angular average in d dimensions
B_1	$\xi(1-\xi)$	$d_\perp\frac{r^i(Y\cdot r)}{r^4} - \frac{d_\perp}{2}\frac{r^i}{r^2}$	$\frac{d_\perp}{d}\frac{Y^i}{r^2} = \frac{Y^i}{r^2}$
B_2	$-\frac{\xi}{1-\xi}$	$-\frac{Y^i}{r^2} + \frac{r^i}{r^2} + \frac{(Y\cdot r)Y^i}{2Y^2r^2}$	$-\frac{Y^i}{r^2}$
B_3	$1-\xi$	$-2\frac{r^i(Y\cdot r)}{r^4}$	$-\frac{2}{d}\frac{Y^i}{r^2}$
B_4	$-\frac{1-\xi}{\xi}$	$-\frac{Y^i}{r^2} - \frac{(Y\cdot r)Y^i}{2Y^2r^2}$	$-\frac{Y^i}{r^2}$
B_5	ξ	$-2\frac{r^i(Y\cdot r)}{r^4} + 2\frac{r^i}{r^2}$	$-\frac{2}{d}\frac{Y^i}{r^2}$
B_6	-1	$-2\frac{r^i(Y\cdot r)}{r^4} + \frac{r^i}{r^2}$	$-\frac{2}{d}\frac{Y^i}{r^2}$

We used $\langle r^i r^j\rangle = \delta^{ij}r^2/d$; terms odd in $\mathbf{r}$ average to zero. Multiplying by the p^+ factors and adding,

$$\xi(1-\xi) + \frac{\xi}{1-\xi} - \frac{2}{d}(1-\xi) + \frac{1-\xi}{\xi} - \frac{2}{d}\xi + \frac{2}{d} = \xi(1-\xi) + \frac{\xi}{1-\xi} + \frac{1-\xi}{\xi}. \tag{3.1}$$

The result does not depend on d:

- the $d_\perp$ of the polarization sum in B_1 cancels the $1/d$ of the angular average;
- the $1/d$ terms of B_3, B_5, B_6 cancel among themselves.

So no $O(\epsilon)$ term of the coefficient multiplies the pole. With $\frac{p^+(k^+-p^+)}{k^{+2}} + \frac{p^+}{k^+-p^+} + \frac{k^+-p^+}{p^+} = \xi(1-\xi) + \frac{\xi}{1-\xi} + \frac{1-\xi}{\xi}$, the integrand of 2b at $x \to z$ is

$$\frac{1}{(2\pi)^3}\frac{g^4}{8\pi^4}\frac{1}{k^+}\frac{dp^+}{2\pi}\,e^{-ik\cdot(w'-z)}\,\frac{1}{k^+}\left[\frac{p^+(k^+-p^+)}{k^{+2}} + \frac{p^+}{k^+-p^+} + \frac{k^+-p^+}{p^+}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2}\,\frac{1}{(x-z)^2}$$
$$\times\Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[N_c U^{af}(y) - N_c U^{af}(z)\Big]\rho^{b'}(x')\rho^f(y) \;+\; \big(\text{zero angular average}\big) + O\Big(\frac{1}{|x-z|}\Big). \tag{3.2}$$

The bracket is the real part of the $g \to gg$ splitting function divided by $2N_c$. In d dimensions every factor $\frac{r^m}{r^2}$ of the wave functions becomes $c_d\frac{r^m}{(r^2)^{d/2}}$ with $c_d = \Gamma(\frac{d}{2})\pi^{1-d/2}$ ($c_2 = 1$), so $\frac{1}{(x-z)^2} \to c_d^2\,[(x-z)^2]^{1-d}$ with the same coefficient.

4 The counterterm

We subtract Eq. (3.2) with $\frac{1}{(x-z)^2}$ replaced by

$$K(z,z,x) - K(y,z,x) = \frac{1}{(x-z)^2} - \frac{(y-x)\cdot(z-x)}{(y-x)^2(z-x)^2} = \frac{(y-x)\cdot(y-z)}{(y-x)^2\,(x-z)^2}. \tag{4.1}$$

The counterterm is therefore

$$\mathrm{CT} = \frac{1}{(2\pi)^3}\frac{g^4}{8\pi^4}\frac{1}{k^+}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,\frac{1}{k^+}\left[\frac{p^+(k^+-p^+)}{k^{+2}} + \frac{p^+}{k^+-p^+} + \frac{k^+-p^+}{p^+}\right]$$
$$\times\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2}\,\big[K(z,z,x) - K(y,z,x)\big]\Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[N_c U^{af}(y) - N_c U^{af}(z)\Big]\rho^{b'}(x')\rho^f(y). \tag{4.2}$$

Then $\frac{d^3N_{2b}}{d^3k} = (2\mathrm{b} - \mathrm{CT}) + \mathrm{CT}$. The first bracket is the rest, Eq. (1.5), and CT is the UV term, Eq. (1.2). The kernel (4.1) has the following properties:

(i) **Same UV limit.** At $x \to z$ it is $\frac{1}{(x-z)^2} + \frac{(y-z)\cdot(x-z)}{(y-z)^2(x-z)^2} + O(1)$, and the second term has zero angular average. So the rest has no $1/(x-z)^2$ term left, for every p^+ in $\Lambda < p^+ < k^+ - \Lambda$.

(ii) **No new IR problem.** At large $|x-z|$ it falls like $|x-z|^{-3}$.

(iii) At $x \to y$ it behaves like $1/|x-y|$, which is integrable.

(iv) **It is a cut at $|x-z| = |y-z|$.** In two dimensions its average over the directions of $x-z$ at fixed $|x-z|$ is exactly

$$\Big\langle\frac{(y-x)\cdot(y-z)}{(y-x)^2(x-z)^2}\Big\rangle_{\text{angle}} = \frac{\theta\big(|y-z|-|x-z|\big)}{(x-z)^2}. \tag{4.3}$$

Proof: in complex notation $\frac{(y-x)}{(y-x)^2} = \frac{1}{\overline{Y}-\bar{r}}$. For $r < |Y|$, expanding in $\bar{r}/\overline{Y}$ gives the average $1/\overline{Y}$, i.e. Y/Y^2. For $r > |Y|$, expanding in $\overline{Y}/\bar{r}$ gives zero.

By (iv), the cut $\frac{\theta(|y-z|-|x-z|)}{(x-z)^2}$, which I gave you before, has the same x integral at every y, z, p^+. So does the smooth form $\frac{1}{(x-z)^2} - \frac{1}{(x-z)^2+(y-z)^2}$: in two dimensions both are $\pi\log\frac{(y-z)^2}{r_0^2}$ with a cutoff $r_0 \to 0$, and in d dimensions they agree with Eq. (1.3) (Table 1). All three choices therefore give the same UV term, and the same rest after the x integral. We use Eq. (4.1) because it is built from the kernels K of your notes, and because it puts the UV term in the same form as your G1-II after F6.

5 The x integral in $d_\perp = 2 - \epsilon$ dimensions

In d dimensions the kernels of Eq. (4.1) become

$$K_d(a,b,x) = c_d^2\,\frac{(a-x)\cdot(b-x)}{[(a-x)^2]^{d/2}\,[(b-x)^2]^{d/2}}, \qquad c_d = \Gamma\Big(\frac{d}{2}\Big)\pi^{1-d/2}, \tag{5.1}$$

because

$$\int\frac{d^dq}{(2\pi)^d}\,e^{iq\cdot r}\,\frac{q^m}{q^2} = \frac{i\,\Gamma(d/2)}{2\pi^{d/2}}\,\frac{r^m}{(r^2)^{d/2}}, \qquad \text{i.e.} \qquad c_d\frac{r^m}{(r^2)^{d/2}} = -2\pi i\int\frac{d^dq}{(2\pi)^d}\,e^{iq\cdot r}\,\frac{q^m}{q^2}. \tag{5.2}$$

Step 1: $K_d(z,z,x)$**.** It equals $c_d^2[(x-z)^2]^{1-d}$, which has no scale, so $\int d^dx\, K_d(z,z,x)=0$. In this integral the UV pole cancels against an IR pole. The IR pole cancels in turn against that of $K_d(y,z,x)$ in step 2, since the difference (4.1) is IR finite.

Step 2: $K_d(y,z,x)$**.** With the Fourier representation of both factors, the x integral gives $(2\pi)^d\delta^{(d)}(q+q')$:

$$\int d^dx\, K_d(y,z,x) = (-2\pi i)^2 \int \frac{d^dq}{(2\pi)^d}\, e^{iq\cdot(y-z)}\, \frac{q\cdot(-q)}{q^2\,q^2} = 4\pi^2 \int \frac{d^dq}{(2\pi)^d}\, \frac{e^{iq\cdot(y-z)}}{q^2} = \pi^{2-d/2}\,\Gamma\Big(\frac{d}{2}-1\Big)\big[(y-z)^2\big]^{1-d/2}. \tag{5.3}$$

(For $d=3$ this is $\pi/|y-z|$, i.e. the known $\int d^3x\,\nabla\frac{1}{|x-y|}\cdot\nabla\frac{1}{|x-z|}=\frac{4\pi}{|y-z|}$.)

Step 3: $\overline{\text{MS}}$**.** With $d=2-\epsilon$ and the factor $\big(\frac{\mu^2e^{\gamma_E}}{4\pi}\big)^{\epsilon/2}$,

$$\begin{aligned}\Big(\frac{\mu^2e^{\gamma_E}}{4\pi}\Big)^{\frac{\epsilon}{2}}\int d^dx\,\big[K_d(z,z,x)-K_d(y,z,x)\big] &= -\pi\,\Gamma\Big(-\frac{\epsilon}{2}\Big)\Big(\frac{(y-z)^2\mu^2e^{\gamma_E}}{4}\Big)^{\frac{\epsilon}{2}}\\ &= \pi\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y-z)^2\mu^2}{4}\right]+O(\epsilon),\end{aligned} \tag{5.4}$$

using $\Gamma(-\frac{\epsilon}{2})=-\frac{2}{\epsilon}-\gamma_E+O(\epsilon)$. This is Eq. (1.3).

Check with the cut. Split $K_d(z,z,x)-K_d(y,z,x)=\theta(R-r)\,c_d^2(r^2)^{1-d}+\big[K_d(z,z,x)-K_d(y,z,x)-\theta(R-r)\,c_d^2(r^2)^{1-d}\big]$, with $R=|y-z|$.

- The first part gives

$$\Big(\frac{\mu^2e^{\gamma_E}}{4\pi}\Big)^{\frac{\epsilon}{2}}c_d^2\,\Omega_d\int_0^R dr\,r^{d-1}r^{2-2d}=\frac{2\pi}{\epsilon}\,\Gamma\Big(1-\frac{\epsilon}{2}\Big)\Big(\frac{R^2\mu^2e^{\gamma_E}}{4}\Big)^{\frac{\epsilon}{2}}=\pi\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{R^2\mu^2}{4}\right]+O(\epsilon), \tag{5.5}$$

  with $\Omega_d=2\pi^{d/2}/\Gamma(d/2)$.

- The second part is UV and IR finite, so it can be done at $d=2$, where it vanishes by Eq. (4.3).

The two results agree. For the smooth form, $\int_0^\infty dt\,\frac{t^{1-d}}{1+t^2}=\frac{\pi}{2\sin(\pi\epsilon/2)}=\frac{1}{\epsilon}+O(\epsilon)$ replaces $\int_0^1 dt\,t^{1-d}=\frac{1}{\epsilon}$, so it also gives the same result.

6 The p^+ integral (exact)

With $\lambda=\Lambda/k^+$:

$$\begin{aligned}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{k^+}\left[\frac{p^+(k^+-p^+)}{k^{+2}}+\frac{p^+}{k^+-p^+}+\frac{k^+-p^+}{p^+}\right] &= \int_\lambda^{1-\lambda}d\xi\Big[\xi(1-\xi)+\frac{1}{1-\xi}+\frac{1}{\xi}-2\Big]\\ &= \Big[\frac{1}{6}-\lambda^2+\frac{2}{3}\lambda^3\Big]+2\log\frac{1-\lambda}{\lambda}-2(1-2\lambda)\\ &= 2\log\frac{k^+-\Lambda}{\Lambda}-\frac{11}{6}+\frac{4\Lambda}{k^+}-\frac{\Lambda^2}{k^{+2}}+\frac{2\Lambda^3}{3k^{+3}}.\end{aligned} \tag{6.1}$$

This is exact; no term is dropped. With the log rule of Sec. 1,

$$2\log\frac{k^+-\Lambda}{\Lambda}=2\log\frac{\vee}{\Lambda}-2\log\frac{\vee}{k^+}+2\log\Big(1-\frac{\Lambda}{k^+}\Big). \tag{6.2}$$

The $-\frac{11}{6}$ comes from $\int_0^1[\xi(1-\xi)-2]\,d\xi$. The log comes from the $\frac{1}{\xi}$ and $\frac{1}{1-\xi}$ parts.

7 The UV term

7.1 Assembling

In Eq. (4.2) we do the x integral with Eq. (5.4) and the p^+ integral with Eq. (6.1). Using $\frac{dp^+}{2\pi}\frac{1}{k^+}=\frac{1}{2\pi}\frac{dp^+}{k^+}$, the prefactor is

$$\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\times\frac{1}{2\pi}\times\pi\times N_c=\frac{1}{(2\pi)^3}\,\frac{g^4N_c}{16\pi^4}\,\frac{1}{k^+}. \tag{7.1}$$

This gives Eq. (1.2). Relative to the LO prefactor $\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2 k^+}$ the factor is $\frac{g^2 N_c}{8\pi^2} = \frac{\alpha_s N_c}{2\pi}$. So, with $L \equiv \big[2\log\frac{\vee}{\Lambda} - 2\log\frac{\vee}{k^+} + 2\log(1-\frac{\Lambda}{k^+}) + \frac{4\Lambda}{k^+} - \frac{\Lambda^2}{k^{+2}} + \frac{2\Lambda^3}{3k^{+3}}\big]$,

$$\left.\frac{d^3N_{2b}}{d^3k}\right|_{\rm UV} = \frac{\alpha_s N_c}{2\pi}\Big[L - \frac{11}{6}\Big] \times \left.\frac{d^3N_{\rm LO}}{d^3k}\right|_{x\to y,\ w\to z} \Big[\text{with } \frac{2}{\epsilon} + 2\gamma_E + \log\frac{(y-z)^2\mu^2}{4} \text{ inside}\Big]. \tag{7.2}$$

The two charges are in the order of your expression, $\rho^{b'}(x')\rho^f(y)$.

7.2 $\overline{\rm MS}$

In $\overline{\rm MS}$ the pole $\frac{2}{\epsilon} = \frac{1}{\epsilon_{\rm std}}$ is subtracted. The γ_E and $\log 4\pi$ are already absorbed in $\mu_{\overline{MS}}$ through the factor $(\mu^2 e^{\gamma_E}/4\pi)^{\epsilon/2}$. This gives Eq. (1.4):

$$\frac{2}{\epsilon} + 2\gamma_E + \log\frac{(y-z)^2\mu^2_{\overline{MS}}}{4} \ \to\ \log\frac{(y-z)^2\mu^2_{\overline{MS}}}{c_0}, \qquad c_0 = 4e^{-2\gamma_E}. \tag{7.3}$$

7.3 The β_0 part and G1-II

The $-\frac{11}{6}$ term is

$$\frac{\alpha_s N_c}{2\pi}\Big(-\frac{11}{6}\Big) = -\frac{\alpha_s\beta_0}{4\pi}, \qquad \left.\frac{d^3N_{2b}}{d^3k}\right|_{{\rm UV},\,\beta_0} = -\frac{\alpha_s\beta_0}{4\pi}\Big[\frac{2}{\epsilon} + 2\gamma_E + \log\frac{(y-z)^2\mu^2}{4}\Big] \times {\rm LO}(y,z). \tag{7.4}$$

Your G1-II (part 1) has

$$\frac{1}{(2\pi)^3}\frac{g^4N_c}{32\pi^4}\frac{1}{k^+}\int e^{-ik(w'-w)}\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\Big[-\frac{2}{\epsilon} - 2\gamma - \log\frac{(x-w)^2\mu^2}{4}\Big]\Big[\frac{11}{3} + \ldots\Big][U(x')-U(w')][U(w)-U(x)]\rho\rho,$$

with the opposite colour sign, $[U(w)-U(x)] = -[U(x)-U(w)]$. Relative to LO its prefactor is $\frac{\alpha_s N_c}{4\pi}$. Per ordering, both rows multiply $\frac{\alpha_s N_c}{4\pi}\big[\frac{2}{\epsilon} + 2\gamma_E + \log\frac{L_\perp^2\mu^2}{4}\big]\times$LO, with $L_\perp = x-w$ in G1-II and $L_\perp = y-z$ in 2b (the same after renaming). The coefficients are:

row	coefficient of $\frac{\alpha_s N_c}{4\pi}\big[\frac{2}{\epsilon} + 2\gamma_E + \log\frac{L_\perp^2\mu^2}{4}\big]\times$ LO
G1-II (part 1)	$+\frac{11}{3} - 2\log\frac{\vee}{\Lambda} + 2\log\frac{\vee}{k^+} + \log\big(1-\frac{k^+}{\vee}\big)$
G1-V, 2b	$-\frac{11}{3} + 4\log\frac{\vee}{\Lambda} - 4\log\frac{\vee}{k^+} + 4\log\big(1-\frac{\Lambda}{k^+}\big) + \frac{8\Lambda}{k^+} - \frac{2\Lambda^2}{k^{+2}} + \frac{4\Lambda^3}{3k^{+3}}$
sum	$2\log\frac{\vee}{\Lambda} - 2\log\frac{\vee}{k^+} + \log\big(1-\frac{k^+}{\vee}\big) + 4\log\big(1-\frac{\Lambda}{k^+}\big) + \frac{8\Lambda}{k^+} - \frac{2\Lambda^2}{k^{+2}} + \frac{4\Lambda^3}{3k^{+3}}$

For G1-II we used $\log(\frac{\vee}{k^+}-1) = \log\frac{\vee}{k^+} + \log(1-\frac{k^+}{\vee})$. The β_0 terms cancel per ordering, including the pole and the μ dependence. So the sum G1-II + 2b needs no coupling renormalization at this order. As noted in `RowV_UV.pdf` (Sec. 6.3), the $\frac{\alpha_s\beta_0}{4\pi}\frac{1}{\epsilon_{\rm std}}$ that the bare coupling of LO requires must then come from another row (the self-energy of the measured gluon after the target); this is still open.

7.4 The pole multiplying the rapidity logarithms

The remaining poles multiply $\log\frac{\vee}{\Lambda}$, $\log\frac{\vee}{k^+}$, $\log(1-\frac{\Lambda}{k^+})$ and the powers of Λ/k^+. They come from the $\frac{1}{\xi}$ and $\frac{1}{1-\xi}$ parts of the splitting function, so they are not a renormalization of the coupling.

- In coordinate space, Eq. (1.3), the $\log\frac{\vee}{\Lambda}$ term is

$$\frac{1}{(2\pi)^3}\frac{g^4N_c}{16\pi^5}\frac{1}{k^+}\,2\log\frac{\vee}{\Lambda}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2}\,\frac{(y-x)\cdot(y-z)}{(y-x)^2(x-z)^2}$$
$$\times\,[U(x')-U(w')]^{ab'}[U(y)-U(z)]^{af}\rho^{b'}(x')\rho^f(y).$$

  This is a two-ρ term with a $1/(x-z)^2$ singularity, like your G1-II after F6.

- The sum of all $\log\frac{\vee}{\Lambda}$ two-ρ terms is JIMWLK⊗LO (`TwoRho_from_final.pdf`, `Evolution_from_final.pdf`). That sum is UV finite: in the Hamiltonian every kernel $K(u,v,t)$ multiplies $[U^\dagger(u)-U^\dagger(t)][U(v)-U(t)]$, which vanishes at $t\to u$ and $t\to v$. So the $1/(x-z)^2$ singularities of the single rows cancel in the sum. With all rows in dimensional regularization, their poles cancel in the same way.

- The UV term plus the rest is your original expression. So the $\log\frac{\vee}{\Lambda}$ content of 2b used in those notes is unchanged.

- The poles that multiply $\log\frac{\vee}{k^+}$, $\log(1-\frac{\Lambda}{k^+})$ and the powers of Λ/k^+ (and $\log(1-\frac{k^+}{\vee})$ in G1-II) must cancel against the other rows in the same scheme. This has not been checked here.

8 The rest

Eq. (1.5) is 2b minus the counterterm (4.2), with the p^+ integral over $\Lambda < p^+ < k^+ - \Lambda$ as in your notes.

- **UV finite.** At $x \to z$ the $1/(x-z)^2$ terms cancel for every p^+: Eq. (3.2) and property (i). What is left has zero angular average or is $O(1/|x-z|)$. So the x integral converges at short distance, and the rest can be evaluated in two dimensions ($d_\perp = 2$).

  Numerically, with a smooth test colour factor and the phase kept:

  - $r^2\langle\text{rest}\rangle \to 0$ at $p^+/k^+ = 0.02,\ 0.37,\ 0.95$;
  - $\int_{r_0<|x-z|<0.3} d^2x$ of the rest converges as $r_0 \to 0$, while the same integral of 2b grows like $2\pi \log(1/r_0)$ (Table 1).

- **Large $|x-z|$.** The rest is 2b there, since the counterterm falls like $|x-z|^{-3}$. As for 2b itself, the x integral converges through the phase $e^{-ik\cdot(z-x)p^+/k^+}$.

- **Rapidity logarithms.** The rest still has the $\frac{k^+}{p^+}$ and $\frac{k^+}{k^+-p^+}$ behaviour at the ends of the p^+ integral, from B_4 and B_2 and from the counterterm. So it still contains $\log\frac{k^+-\Lambda}{\Lambda}$ terms, now with UV-finite kernels. No plus prescription is applied; the p^+ integral is over $\Lambda < p^+ < k^+ - \Lambda$.

- **The c.c.** is the complex conjugate of Eqs. (1.2) and (1.5); it reverses the order of the charges.

9 Checks

All numbers come from `part2b_checks.py` (`part2b_checks.json`). In the checks $z = 0$, $y = (0.8, -0.5)$, $\mathbf{k} = (1.3, 0.4)$. The colour factor $N_c[U(y) - U(x)]$ is replaced by a smooth test function $G(y) - G(x)$, and the counterterm gets $G(y) - G(z)$.

check	result
short distance: $r^2\langle\text{integrand}\rangle\,\frac{(y-z)^2}{(y-z)^i}$ vs. $\xi(1-\xi) + \frac{\xi}{1-\xi} + \frac{1-\xi}{\xi}$; $d = 2$ and $d = 3$ ($d_\perp = d$); $\xi = 0.1, 0.37, 0.8$; $r = 10^{-4}$	max relative difference 1.6×10^{-8}
Eq. (4.3): $r^2\langle K(z,z,x) - K(y,z,x)\rangle$ at $\lvert x-z\rvert/\lvert y-z\rvert = 0.01, 0.32, 0.90, 1.10, 3.18, 31.80$	1.000000, 1.000000, 1.000000, 0.000000, 0.000000, 0.000000
$\xi = 0.02$: $r^2\lvert\langle\text{2b}\rangle\rvert$ and $r^2\lvert\langle\text{rest}\rangle\rvert$ at $r = 10^{-2}, 10^{-3}, 10^{-4}$	16.7232, 16.7240, 16.7240; 8.6×10^{-4}, 8.6×10^{-6}, 8.6×10^{-8}
$\xi = 0.37$: $r^2\lvert\langle\text{2b}\rangle\rvert$ and $r^2\lvert\langle\text{rest}\rangle\rvert$ at $r = 10^{-2}, 10^{-3}, 10^{-4}$	0.8604, 0.8604, 0.8604; 5.9×10^{-5}, 5.9×10^{-7}, 5.9×10^{-9}
$\xi = 0.95$: $r^2\lvert\langle\text{2b}\rangle\rvert$ and $r^2\lvert\langle\text{rest}\rangle\rvert$ at $r = 10^{-2}, 10^{-3}, 10^{-4}$	6.5127, 6.5137, 6.5137; 1.1×10^{-3}, 1.1×10^{-5}, 1.1×10^{-7}
$\xi = 0.2$: $\mathrm{Re}\int_{r_0<\lvert x-z\rvert<0.3} d^2x$ (component $i = 1$), $r_0 = 10^{-2}, \ldots, 10^{-5}$: rest; 2b	0.250416, 0.250689, 0.250692, 0.250692; −31.89, −53.65, −75.41, −97.16
$\xi = 0.7$: $\mathrm{Re}\int_{r_0<\lvert x-z\rvert<0.3} d^2x$ (component $i = 1$), $r_0 = 10^{-2}, \ldots, 10^{-5}$: rest; 2b	0.300763, 0.301098, 0.301102, 0.301102; −21.36, −36.02, −50.68, −65.35
same: 2b minus $\pi \log\frac{0.09}{r_0^2}\times$(coefficient of Eq. (3.2)) minus rest	max 2.5×10^{-14}
$\xi = 0.2$, $\lvert x-z\rvert = 10, 10^2, 10^3$: $\lvert x-z\rvert^3 \max\lvert\text{CT}\rvert$; $\lvert x-z\rvert^2 \max\lvert\text{2b}\rvert$; $\lvert x-z\rvert^2\lvert\langle\text{2b}\rangle\rvert$ without phase	1.567, 1.432, 1.420; 5.186, 4.956, 4.920; 2.287, 2.303, 2.303
$\xi = 0.7$, $\lvert x-z\rvert = 10, 10^2, 10^3$: $\lvert x-z\rvert^3 \max\lvert\text{CT}\rvert$; $\lvert x-z\rvert^2 \max\lvert\text{2b}\rvert$; $\lvert x-z\rvert^2\lvert\langle\text{2b}\rangle\rvert$ without phase	1.056, 0.965, 0.957; 2.624, 3.736, 3.627; 1.165, 1.166, 1.166
$\xi = 0.2$: $\mathrm{Re}\int_{1.5<\lvert x-z\rvert<R_{\max}} d^2x$ rest (phase kept), $R_{\max} = 25, 50, 100, 200$	6.785, 6.981, 7.001, 6.961
$\xi = 0.7$: $\mathrm{Re}\int_{1.5<\lvert x-z\rvert<R_{\max}} d^2x$ rest (phase kept), $R_{\max} = 25, 50, 100, 200$	−1.213, −1.171, −1.182, −1.181
$d_\perp = 2-\epsilon$ integrals minus $\pi[\frac{2}{\epsilon} + 2\gamma_E + \log\frac{R^2\mu^2}{4}]$: K form (Eq. (5.4)), cut, smooth form (`sympy`)	0; 0; 0
K-form formula at $d = 3$ minus $\frac{1}{4}\cdot\frac{4\pi}{R}$ (`sympy`)	0
Eq. (6.1): exact integral minus the right-hand side at $\lambda = \frac{1}{3}, \frac{1}{10}, \frac{1}{1000}$ (`sympy`, 30 digits)	5.4×10^{-166}
prefactor of the UV term/$\frac{1}{(2\pi)^3}\frac{g^4 N_c}{16\pi^4}\frac{1}{k^+}$; /$\left(\frac{\alpha_s N_c}{2\pi}\times\text{LO prefactor}\right)$	1; 1
$\frac{\alpha_s N_c}{2\pi}(-\frac{11}{6}) + \frac{\alpha_s \beta_0}{4\pi}$; β_0 coefficient G1-II(1) + 2b	0; 0

Table 1: Numerical and symbolic checks.